

Research Workflow Manual

BLG + β -Casein Adsorption Project — Mac Mini Research Station

Chalakon Pornjariyawatch — COMFHA, Kasetsart University

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Why this manual exists

This project accumulated 8 slash-commands over months of work, then 6 more were added in a single session (2026-09-04) to close real gaps: supervisor feedback was scattering across 3 different files with no single source of truth, session reports were being sent to P.P. under the wrong filename convention, there was no dedicated review role independent of Claude’s own framing, and Mac Mini → MacBook syncs were being done ad hoc by hand. This manual is the map: what each piece is for, how they fit together, and the order you actually use them in.

The 5-role structure this project’s workflow is organized around:

1. **Gap-finder** — literature landscape scanning, spark next research directions (not yet built — intentionally last, gated on the CAS side of the project landing)
2. **P.P. feedback tracker** — supervisor feedback, collected and applied
3. **Research execution + personal report** — GROMACS/Python work, with a technical log for yourself, separate from what goes to P.P.
4. **Teach analysis + figures + VMD** — you build these yourself (`scripts/analysis/*.py`, `scripts/figures/*.py`, and, added later the same day, `scripts/figures/vmd/*.tcl` molecular renders); Claude teaches concepts and reviews, doesn’t write it for you
5. **Fable advisor** — an independent, professor-style review checkpoint

Role 4 pilot targets identified, not yet started: `scripts/figures/blg_fig2.dynamics.py` (matplotlib — all 4 BLG RMSD/RMSF/ R_g datasets ready) and, as a candidate, before/after VMD calyx renders bracketing R1’s long contact event (362–419.5 ns) — ties directly to the real RMSD finding rather than being a generic exercise.

1 The Command Reference

1.1 Session lifecycle — the commands that bookend every session

Command	What it does
<code>/start-research</code>	Full session init: identity/deviation self-check, reads project brief + memory, health-checks GROMACS/Python/git/disk/RAM, checks both clusters’ job queues, prints open P.P. feedback items with any near-term deadline flagged , and recommends one concrete next action. Run this first, every session.

Command	What it does
/journey	Research journey board — every milestone (✓/→/○) derived from actual files/git/cluster state, never from memory alone — plus a narrative PDF recap of <i>this session</i> to <code>progress-reports/session_report_*.pdf</code> . This is your personal, technical log : every command run, every mistake caught, what to double-check next time. Use at the end of a work session.
/end-session	Saves memory, updates <code>docs/PP_FEEDBACK_LOG.md</code> with anything new from the session, verifies git is clean, writes the cross-machine “Last Session” handoff into <code>CLAUDE.md</code> , commits and pushes. Run this last.
/self-check	Full identity/integrity/deviation diagnostic, Connor-style. Run whenever something feels off, or on request.

Standard session shape: `/start-research` → work → `/journey` → `/end-session`.

1.2 Supervisor (P.P.) management — built 2026-09-04, Role 2 & 3

Command	What it does
/pp-feedback	The front door to <code>docs/PP_FEEDBACK_LOG.md</code> , the canonical record of everything P.P. has said, decided, or left open. Three modes: <code>/pp-feedback</code> (no args) — print all open items, flag anything with a deadline under 30 days <code>/pp-feedback log "..."</code> — record new feedback (a meeting, a LINE message, an email); always shown to you before it's written <code>/pp-feedback apply</code> — cross-check current work against open items, flags what's been addressed but not yet marked resolved (never auto-resolves — always asks)
/pp-report [BLG CAS all]	Generates the report that actually goes <i>to</i> P.P. — results and open questions only, numbers read live from <code>results/analysis/*.npz</code> every time (never hard-coded, never copied from an old PDF). Output: <code>progress-reports/for_PP_<date>.pdf</code> .

Naming rule (fixed 2026-09-04, don't break it): `session_report_*.pdf` = personal (`/journey`). `for_PP_*.pdf` = supervisor-facing (`/pp-report`). Before this split, session reports and P.P. reports used the same filename pattern and got mixed up at least once.

1.3 Research execution — GROMACS & cluster work

Command	What it does
/new-sim <protein> <sim-type> <name> <cluster>	Scaffold a new MD simulation — local directory structure + matching cluster run directory + SLURM script. Example: /new-sim BLG production replica4 ku-cluster.
/submit-job <cluster> <protein> <sim-type> <tpv-name>	Prepare and submit a GROMACS job. Always shows the full sbatch script and waits for explicit “yes” before submitting — this is a hard project rule, not optional.
/sync-results <run>	Sync simulation output from a cluster back to the Mac Mini. Always dry-runs the rsync first and shows exactly what will transfer before doing it for real.
/sync-macbook <what>	Sync docs/reports from the Mac Mini to the MacBook Air (for quick viewing, not a replacement for the git push/pull cycle). Always tries macbook-ts (Tailscale) first — the LAN alias (macbook) has drifted via DHCP and timed out twice in this project’s history (2026-07-22, 2026-09-04); dry-runs the rsync first, same as /sync-results.
/check-cluster	Live queue from both ku-cluster and Nontri HPC (ku-ai), no other steps.
/analyze <type>	Run an analysis script — gate, rmsd, rmsf, sasa, contact, energy, orientation, fig2, fig3, fig4. Always activates research-env first.

1.4 Quality & review

Command	What it does
/paper-review [focus]	Structured review of main.tex against JCIS submission standards. Optional focus: citations, science, latex, checklist.

Command	What it does
<code>/fable-review</code> <code><what></code>	Role 5. Spawns a genuinely <i>fresh</i> agent (never a fork) on the Fable model to independently review a decision, a plan, or a piece of work — no inherited context, no pre-digested framing from Claude, forms its own view via its own file reads. Every real bug this project has caught via Fable (phosphoserine charge state, a stale-tpz bug, the SASA-normalization gap) came from exactly this zero-context independence. Always cross-checked before being reported as fact — Fable has one confirmed false alarm on record in this project (a “biggest risk” flag on something already resolved), so its findings are leads to verify, not conclusions to relay directly.

2 How a typical session actually flows

```

/start-research
|
+-- health check (GROMACS, Python, git, disk, RAM)
+-- both clusters' job queues
+-- P.P. open items (from docs/PP_FEEDBACK_LOG.md) -- flagged if a deadline is close
|
v
DO THE ACTUAL WORK
|
+-- research/GROMACS work --> /new-sim, /submit-job, /sync-results, /check-cluster
+-- analysis work          --> /analyze (or, per Role 4, you write it, Claude teaches)
+-- new P.P. feedback?     --> /pp-feedback log "..." (log it same day, don't wait)
+-- need a second opinion?--> /fable-review "..." (fresh, independent, verified)
|
v
/journey                                /pp-report (only when a report to P.P. is due --
|                                         not every session)
+-- milestone board
+-- session_report_*.pdf
    (your personal, technical log)
|
v
/end-session
|
+-- memory updated
+-- docs/PP_FEEDBACK_LOG.md updated with anything new
+-- CLAUDE.md "Last Session" handoff rewritten
+-- git commit + push

```

3 Where things actually live

Location	What's there
CLAUDE.md	The authoritative project brief. “Last Session” section at the top is the cross-machine handoff (Mac Mini ↔ MacBook). Read every session.
docs/PP_FEEDBACK_LOG.md	Canonical supervisor feedback — not CLAUDE.md, not memory. Those may summarize it; this file is the source of truth.
docs/ paper1_expansion_plan.md	The original, fuller capture of P.P.’s June 9 meeting — figure plans, raw notes, open decisions with drafted recommendations.
docs/METHODS.md, docs/COMFHA_Science_Notes.md	Science/methods notes.
results/analysis/*.npz, results/gate_analysis/*.npz	Every analysis result, as actual numbers — the single source of truth for any number quoted anywhere (paper, reports, this manual). Never trust a remembered number over these files.
progress-reports/ session_report_*.pdf	Personal log, from /journey.
progress-reports/ for_PP_*.pdf	Supervisor-facing report, from /pp-report.
Claude’s private memory (not in this repo)	Session continuity, technical gotchas, your working preferences — Claude reads this automatically every session; you generally don’t need to touch it directly.

4 Quick reference

Starting a session	/start-research
Ending a session	/journey then /end-session
Checking what P.P. is waiting on	/pp-feedback
Logging new feedback from P.P.	/pp-feedback log "..."
Sending P.P. an update	/pp-report
Starting a new simulation	/new-sim ...
Running an analysis	/analyze <type>
Second opinion on a decision	/fable-review "..."
Checking paper readiness	/paper-review
Something feels wrong	/self-check

Generated 2026-09-04 as part of building this project’s 5-role workflow structure. Update this manual whenever a command is added, renamed, or its behavior changes materially — a stale manual is worse than no manual.